Sports Medicine
https://doi.org/10.1007/s40279-021-01538-2

ORIGINAL RESEARCH ARTICLE

Muscle Fibre Typology as a Novel Risk Factor for Hamstring Strain Injuries in Professional Football (Soccer): A Prospective Cohort Study

E. Lievens[1] · K. Van Vossel[1] · F. Van de Casteele[1] · E. Wezenbeek[2] · D. Deprez[3] · S. Matthys[1,4] · B. De Winne[5] · S. McNally[6] · W. De Graaf[7] · J. B. Murdoch[8] · J. G. Bourgois[1,4] · E. Witvrouw[2] · Wim Derave[1]

Accepted: 5 August 2021
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2021

Abstract

Background Hamstring strain injuries (HSI) are prevalent in team sports and occur frequently in the later phase of matches. In the search for interindividual factors that determine muscle fatigue and possibly injury risk, muscle fibre typology is a likely candidate.

Objective The aim of the study was to determine whether muscle fibre typology is a risk factor for HSI.

Methods A prospective cohort study was conducted over three seasons in professional football players competing in the Belgian Jupiler Pro League ($n=118$) and in the English Premier League ($n=47$). A total of 27 HSI were sustained during this period. Muscle fibre typology was non-invasively estimated using proton magnetic resonance spectroscopy and was characterized as a fast, slow, or intermediate typology based on the carnosine concentration in the soleus. A multivariate Cox model was used to identify risk factors for HSI.

Results Football players exhibited a wide variety of muscle typologies (slow 44.9%, intermediate 39.8%, fast 15.3%). In the combined cohort, players with a fast typology displayed a 5.3-fold (95% confidence interval [CI] 1.92–14.8; $P=0.001$) higher risk of sustaining an index HSI than slow typology players. This was also independently observed in both leagues separately as, respectively, a 6.7-fold (95% CI 1.3–34.1; $P=0.023$) and a 5.1-fold (95% CI 1.2–20.4; $P=0.023$) higher chance was found in fast typology players than in slow typology players of the Jupiler Pro League and the Premier League cohort.

Conclusion We identified muscle fibre typology as a novel and potent risk factor for HSI in team sports.

1 Introduction

Hamstring strain injuries (HSI) are the most prevalent non-contact injuries in many team sports, including football in which they are responsible for 12% of all injuries [1]. Given

Key Points

We identified muscle fibre typology as an important risk factor for hamstring strain injury (HSI) in football, with fast typology players displaying a 5.3-fold higher risk of experiencing a new HSI than slow typology players.

In the continous analysis, a hazard ratio of 1.83 was found (95% confidence interval [CI] 1.25–2.66; $P=0.002$), which means that when the Z-score increases by one unit (indicating a faster typology), an athlete will have an 83% higher chance of a HSI.

Muscle fibre typology was non-invasively estimated by proton magnetic resonance spectroscopy, accommodating its use in elite sport.

✉ Wim Derave
Derave@ugent.be

1 Department of Movement and Sports Sciences, Ghent University, Watersportlaan 2, 9000 Ghent, Belgium

2 Department of Rehabilitation Sciences, Ghent University, Ghent, Belgium

3 Club Brugge KV, Brugge, Belgium

4 Koninklijke Atletiek Associatie Gent, Ghent, Belgium

5 Zulte Waregem, Waregem, Belgium

6 Manchester United FC, Manchester, UK

7 Canon Medical Systems Europe B.V., Zoetermeer, The Netherlands

8 Canon Medical Research, Cleveland, OH, USA

Published online: 13 September 2021

the high incidence and the recurrence rate (16%, characterized by an even longer rehabilitation), these injuries have a major impact on both sport performance and sport medical costs [1–3]. A number of risk factors for HSI have been characterised, including increasing age, previous hamstring injury, lower eccentric hamstring strength, and shorter fascicle length [4, 5]. Despite decades of intensive scientific attention, the incidence of HSI is not declining [3], which could indicate that the nature of the problem is insufficiently understood or that compliance with current prevention strategies is insufficient. On the other hand, high-intensity running distances during football games have markedly increased over the years [6], but this has not led to an associated increased HSI incidence, which could indicate at least partial success in knowledge translation.

HSI are more frequently sustained in the later phase of matches, highlighting muscle fatigue as a potential contributing factor to injury susceptibility [1, 7]. Although evidence for an association between fatigue and hamstring injury is principally epidemiological, a number of fatigue-induced alterations could theoretically increase susceptibility to injury [8]: altered muscle activation and neuromuscular coordination [9, 10], altered pelvic position [10], reduction in proprioceptive ability [11, 12], decreased lower limb muscle stiffness [13], and reduced eccentric hamstring strength [5, 10, 14–16]. In the search for interindividual factors that determine increased muscle fatigue—and thus possibly HSI risk—muscle fibre type composition is a likely candidate since fast-twitch muscle fibres are less resistant to fatigue than slow-twitch fibres [17]. It can therefore be hypothesized that players with a predominant fast typology (FT) exhibit more fatigue during matches and are, as such, at increased risk for HSI.

Direct evidence of dominant fast-twitch (or slow-twitch) fibre composition being a risk factor for muscle strain injuries in vivo is currently lacking. This is probably because the gold standard of measuring muscle fibre type composition, i.e. a histological evaluation of a muscle needle biopsy, has significant limitations and is not feasible in a large cohort of professional (football) players. Our research group has developed a novel non-invasive approach to estimate muscle fibre typology through the measurement of muscle carnosine, a characteristic metabolite of fast-twitch fibres, by proton magnetic resonance spectroscopy (^{1}H-MRS) [18]. More precisely, the muscle typology was estimated in the soleus, as this muscle was found to be the most reliable during test–retest [19], exhibited the lowest biological variability [20], and allowed comparison of the values of these football players against a large reference population of healthy, non-specifically trained individuals. Moreover, because of the presence of an across-muscle phenotype, implying that the fibre type proportion of one muscle is indicative of the proportion in the body as a whole, the muscle fibre typology of the hamstrings will be related to the typology in the soleus [21, 22]. This approach therefore enables investigation of whether muscle fibre typology is a risk factor for muscle strain injury.

This study aimed to explore the muscle fibre typology of professional football players (research question 1). As professional football requires both endurance capacity and sprint ability [6], it was hypothesised that football teams would encompass a heterogeneous group of muscle typologies, which would allow us to assess muscle fibre typology as a risk factor for HSI incidence (research question 2). The 1- to 3-year follow-up was conducted in 165 adult professional football players. Moreover, since this study included two different cohorts (the Jupiler Pro League in Belgium and the Premier League in the UK), we also examined whether the findings were preserved when analysing the two professional leagues individually and separately. We hypothesised that a faster typology would be linked to a higher risk for HSI.

2 Methods

2.1 Study Design

This study was conducted over three consecutive seasons (2015–2018) in four teams of the Belgian Jupiler Pro League and one team of the Premier League. Throughout this period, the muscle fibre typology of 165 individual adult professional football players (Jupiler Pro League: $n = 118$; Premier League: $n = 47$) was non-invasively determined. The Jupiler Pro League players were used to determine the muscle typology of football players relative to a reference population (research question 1; Fig. 1). The HSI risk assessment included players of both leagues. Moreover, players were excluded if they were goal keepers, if they had a HSI in the 6 months before the screening (to avoid investigating re-injuries), if they left the team before the minimum follow-up of 1 year was reached, if the injury follow-up data were either not provided (two players in the Premier League) or were not provided with enough accuracy (one in the Jupiler Pro League team). After applying these exclusion criteria, 61 Jupiler Pro League players (age 26.7 ± 4.79 years; weight 76.2 ± 7.15 kg; height 181 ± 7.18 cm; median duration of exposure 471 days [interquartile range 360]) and 34 Premier League players (age 22.2 ± 1.76 years; weight 72.4 ± 6.68 kg; height 181 ± 6.38 cm; median duration of exposure 510 days [interquartile range 411]) were included in the HSI risk study.

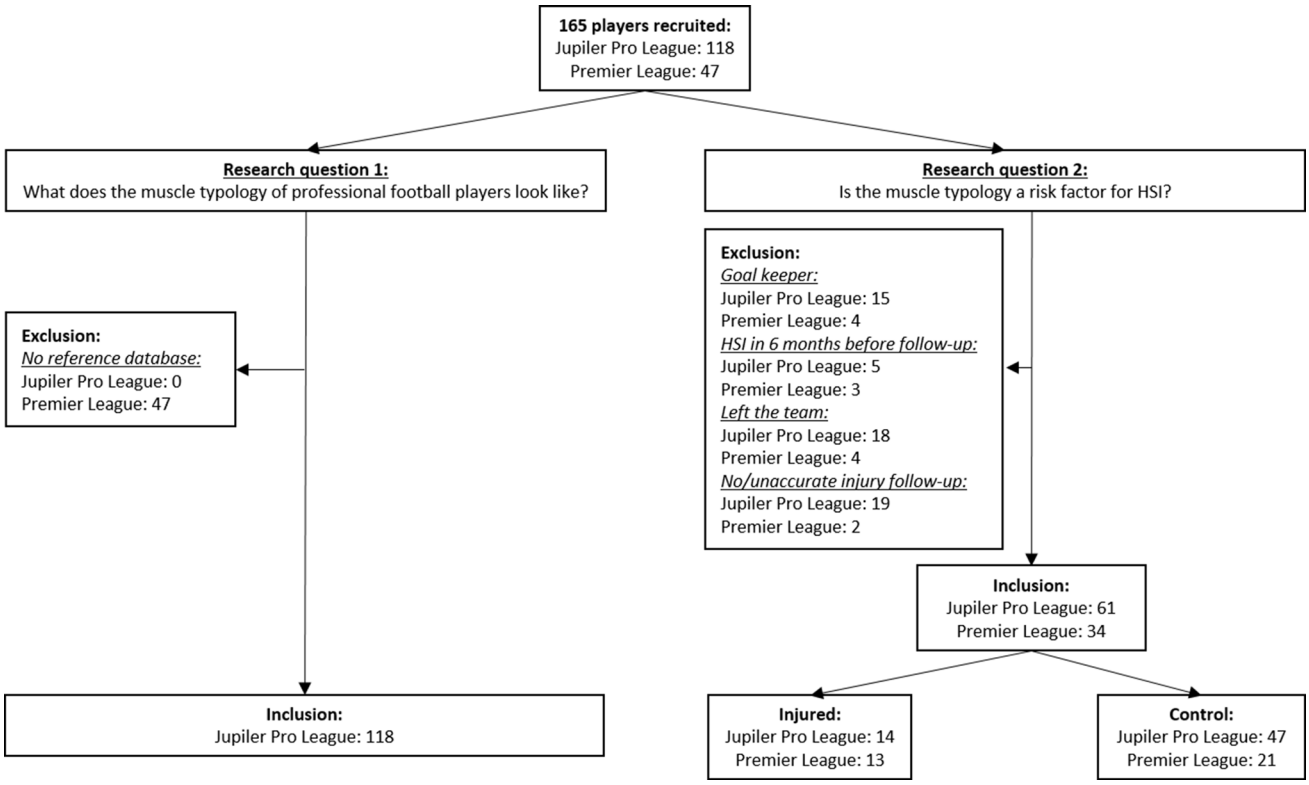

Fig. 1 Flowchart. *HSI* hamstring strain injury

2.2 Muscle Fibre Typology Assessment

Muscle carnosine content was measured using ^{1}H-MRS in the right soleus muscle on a 3 T whole-body magnetic resonance imaging (MRI) scanner (Jupiler Pro League: Trio, Siemens; Premier League: Vantage Titan, Canon), based on the size of the carnosine peak at 8 ppm. Subjects lay in a supine position, and the lower leg was fixed in a knee coil (Jupiler Pro League) or a flexible wrap coil (Premier League). A single-voxel point-resolved spectroscopy (PRESS) sequence with the following parameters was used for the carnosine scan: repetition time of 2000 ms, echo time of 30 ms (Jupiler Pro League)/32 ms (Premier League), 128 excitations, 1024 data points, spectral bandwidth of 1200 Hz (Jupiler Pro League)/2604 Hz (Premier League), acquisition time of 4.27 min, and a voxel size of 40 × 12 × 30 mm. A representative spectrum is displayed in Fig. 2.

The absolute carnosine content (mM) in the Jupiler Pro League was calculated as described by Baguet et al. [18] using the phantom replacement technique. We previously used this technique (albeit on different test groups) to determine the methodological and biological variability of the muscle carnosine content in the soleus. The methodological test–retest variation within the same day was 4.3% ($n = 9$) [19], and the biological variation over 15 weeks, 5–6 months, and 1, 2, and 3 years was, respectively, 9.8% ($n = 7$) [20], 6.5% ($n = 17$), 6.9% ($n = 11$), 10.5% ($n = 7$), and 8.8% ($n = 6$) (unpublished observations). These values suggested that an increase in the time interval did not change the variation of the measurement.

In the Premier League, the carnosine content (mM) was calculated using unsuppressed water (eight excitations) as an internal reference. The formula used for this internal reference technique was as follows:

Carnosine concentration (mM) = [(Signal$_{carnosine}$/Signal$_{water}$)] × [(Correction factor T1$_{water}$)/(Correction factor T1$_{carnosine}$)] × [(Correction factor T2$_{water}$)/(Correction factor T2$_{carnosine}$)] × (Concentration of water in the muscle) × number of protons in water.

Carnosine concentration (mM) = [(Signal$_{carnosine}$/Signal$_{water}$)] × [(1 − exp(− 2000/1340))/(1 − exp(− 2000/1488)) × [(exp(− 32/28.7))/(exp(− 32/152))] × (0.7 × 55000) × 2.

The relaxation correction factors for carnosine were described by Baguet et al. [23] and the correction factors for water by MacMillan et al. [24]. The concentration of water in a muscle was deduced from the molar concentration of water (55,000 mM) and the approximate water content of skeletal muscle tissue (0.7 L/kg wet weight of tissue) [25]. The methodological test–retest variation using this technique was 4.0% (unpublished data; $n = 3$). Moreover,

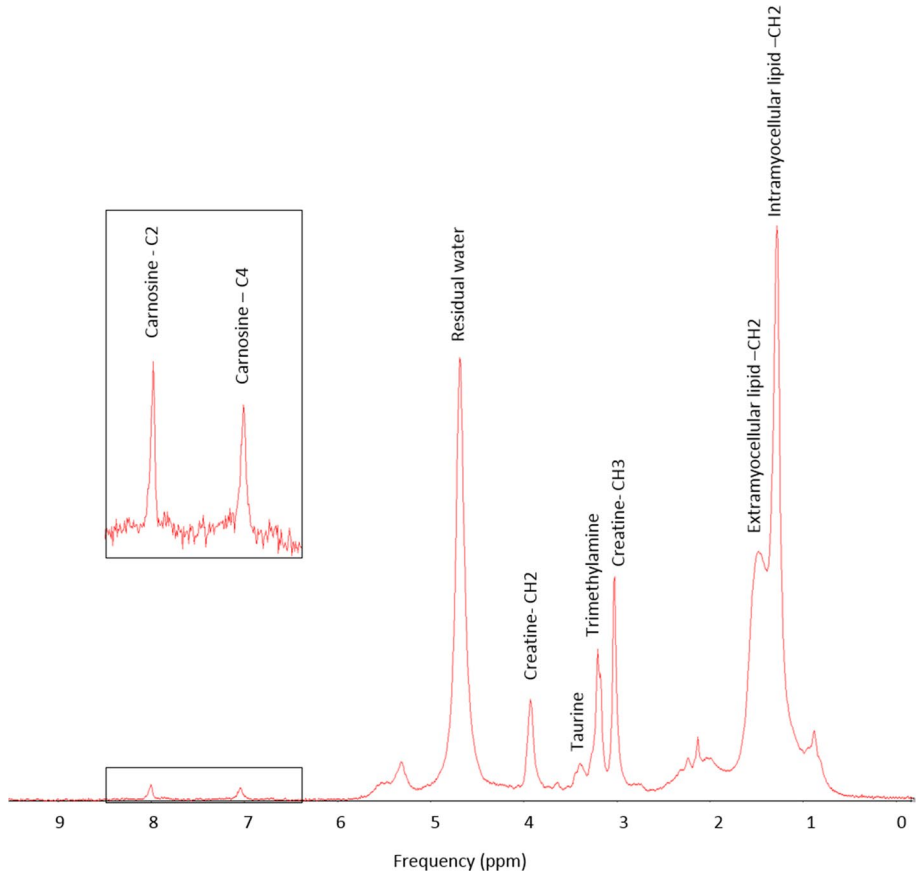

Fig. 2 Representative proton magnetic resonance spectrum of the soleus

a cross-vendor comparison performed on a Siemens and a Canon 3 T scanner ($n=7$) indicated a good correlation ($R^2=0.862$) between soleus carnosine concentrations with both devices [26].

In the Jupiler Pro League, the carnosine concentrations were converted into *Z*-scores relative to an age- and sex-matched control population of 98 active, healthy non-specifically trained individuals. This allowed us to compare the *Z*-scores of the football players with those of the general population. A negative *Z*-score indicates a slower muscle fibre typology when compared with the general population, whereas a positive *Z*-score indicates a faster muscle fibre typology. In addition, the *Z*-scores of the football players were compared with those of elite athletes in typical sprint-type sports ($n=15$; 100–200 m running, BMX and track-sprint cycling disciplines) and endurance-type sports ($n=61$; road cycling, mountain bike, triathlon, and 1500 m to marathon running), which were scanned on the same device.

In the Premier League, the carnosine concentrations were converted into a *Z*-score relative to all male football players scanned on the same device ($n=65$), as we did not have an independent reference database on a Canon system. For both cohorts, football players were assigned into three groups based on their *Z*-score: a slow typology (ST) group with a *Z*-score less than or equal to -0.5, an intermediate typology (IT) group with a *Z*-score between -0.5 and 0.5, and a FT group, with a *Z*-score ≥ 0.5.

2.3 Hamstring Strain Injury (HSI) Assessment

Prior to the scan, personal data (age, height, weight, position) were provided to the researchers by the medical staff. During the follow-up period, the type and duration of all injuries were documented. The exposure for each player was determined by subtracting the injured period (days) from the overall follow-up period (days). The endpoint of the follow-up period was either the end of season 2017–2018 or the day that the player left the team or sustained a HSI. A HSI was defined as acute pain in the posterior thigh that occurred during training or match, which resulted in the inability to participate in the next training session or match, and was later diagnosed by club medical staff. In two of the Jupiler Pro League clubs, a hamstring strain was diagnosed clinically with the support of echography. Depending on the size and the location of the injury, an MRI was consulted for a more in-depth diagnosis. In one Jupiler Pro League team and the Premier League team, more than 95% of the hamstring stains were confirmed with MRI.

2.4 Statistics

2.4.1 Muscle Fibre Typology Assessment

A one-way analysis of variance (ANOVA) was used to determine whether the muscle fibre typology *Z*-scores were different between playing positions in both leagues separately. The following assumptions for a one-way ANOVA were determined. First, the normality of the data was assessed in the four different position groups both visually and using the Shapiro–Wilk normality test. The latter was significant in defenders of the Premier League. However visual assessment showed a positive skewness, which was not interpreted as problematic. Second, the homogeneity of the data was assessed using Levene's tests, results of which were non-significant. Finally, samples were independent in the Jupiler Pro League and the Premier League.

2.4.2 HSI Risk Assessment

A multivariable Cox model was used to determine the relation between the muscle fibre typology (*Z*-score based on carnosine content) and the sustained HSI. The following assumptions of the multivariable Cox model were determined: (1) proportional hazard assumption was assessed by a log minus log analysis and was satisfied for the Jupiler Pro League, the Premier League, and the combined cohort; (2) the linearity of continuous covariates was assessed using a Martingale residuals scatterplot; (3) samples were found to be independent in the Jupiler Pro League and the Premier League (in the combined cohort, league was added as a covariate); and (4) censoring was found to be non-informative.

We investigated whether correction was needed for anthropometric variables, position, age, and league. First, a directed acyclic graph was compiled, showing that correction might be needed for age and league. Next, univariate Cox model analyses were performed, allowing reductions of the number of variables, since variables were only included for further analysis if $P < 0.2$. Second, multivariate Cox model analyses were performed to identify significant contributors to the development of HSI. To calculate the receiver operating characteristic (ROC) curves, an equal time of exposure for all players was a prerequisite, so only the follow-up from the first year was used. The area under the curve (AUC) and the specificity and sensitivity of the model were determined.

3 Results

3.1 Muscle Fibre Typology Assessment

Players from the Jupiler Pro League exhibited a much wider variety of muscle typologies than elite athletes from individual sport disciplines. In contrast to the sprint and endurance sports, football players were more evenly distributed in all three main categories (ST 44.9%, IT 39.8%, FT 15.3%). These muscle fibre typology distributions did not differ between playing positions in the Jupiler Pro League or in the Premier League (Fig. 3; $P = 0.432$ and $P = 0.798$, respectively).

3.2 HSI Risk Assessment

In the combined cohort, a total of 27 HSI were reported during the follow-up period. When correcting for age and league, the muscle fibre typology Z-score showed a hazard ratio (HR) of 1.83 (95% confidence interval [CI] 1.25–2.66; $P = 0.002$; Table 1). This means that when the Z-score increases by one unit, an athlete would have a 83% higher chance of a HSI. In the Jupiler Pro League (14 HSI), a HR

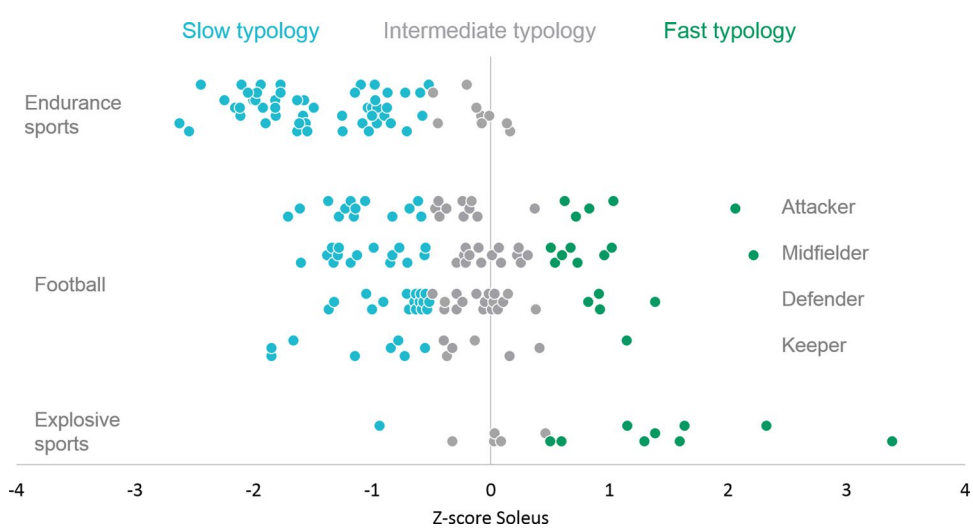

Fig. 3 Muscle fibre typology of Jupiler Pro League players compared with athletes from individual sport disciplines. Values are expressed as *Z*-scores. Football is further subdivided into the different field positions

Table 1 Multivariable Cox model and receiver operating characteristic analysis

Cohort	Variable	Multivariable Cox model (1–3 year)				ROC analysis (1 year)			
		HR	95% CI	P value	HSI (*n*)	AUC	95% CI	Sensitivity/specificity	HSI (*n*)
Combined cohort	*Z*-score soleus	1.83	1.25–2.66	**0.002**	27	0.669	0.501–0.838	0.714/0.593	14
	Age	1.07	0.948–1.200	0.283					
	League	1.80	0.688–4.69	0.232					
Jupiler Pro league	*Z*-score soleus	2.05	1.15–3.66	**0.015**	14	0.752	0.531–0.974	0.750/0.717	8
	Age	1.13	0.996–1.29	0.057					
Premier league	*Z*-score soleus	1.81	1.07–3.05	**0.026**	13	0.804	0.636–0.971	0.833/0.607	6
	Age	0.552	0.310–0.981	**0.043**					

The bolded *P* values indicate a statistically significant factor in the analysis

CI confidence interval, *HR* hazard ratio, *HSI* hamstring strain injury

of 2.05 (95% CI 1.15–3.66; $P = 0.015$) was found when correcting for age. Moreover, these findings were mirrored in the Premier League (13 HSI): HR 1.81 (95% CI 1.07–3.05; $P = 0.026$).

When dividing the players into ST, IT, and FT categories (as shown in Fig. 4), the HSI incidence in the combined cohort, the Jupiler Pro League, and the Premier League was, respectively, 5.32-fold (95% CI 1.92–14.8; $P = 0.001$), 6.66-fold (95% CI 1.30–34.1; $P = 0.023$), and 5.05-fold (95% CI 1.24–20.4; $P = 0.023$) higher in the FT than in the ST players.

ROC curve analysis of the variables muscle fibre typology, age, and league (Table 1) revealed an AUC of 0.669, 0.752, and 0.804 in, respectively, the combined cohort, Jupiler Pro League, and the Premier League, indicating poor to good combined sensitivity and specificity.

4 Discussion

The present study demonstrated heterogeneous muscle typologies in a large cohort of football players. Some players had muscles resembling those of endurance athletes, some were intermediate, and some resembled explosive sprint-type athletes. The clear heterogeneity in the present study allowed us to investigate muscle fibre typology as a potential risk factor for HSI.

The main finding of the present study was that the muscle fibre typology was strongly related to the risk for HSI. Football players with a FT displayed a 5.3-fold higher risk than ST players in sustaining an index HSI, and this was independently confirmed in two different leagues (Jupiler Pro League and Premier League). Muscle fibre typology seems to be an important risk factor, as these values were

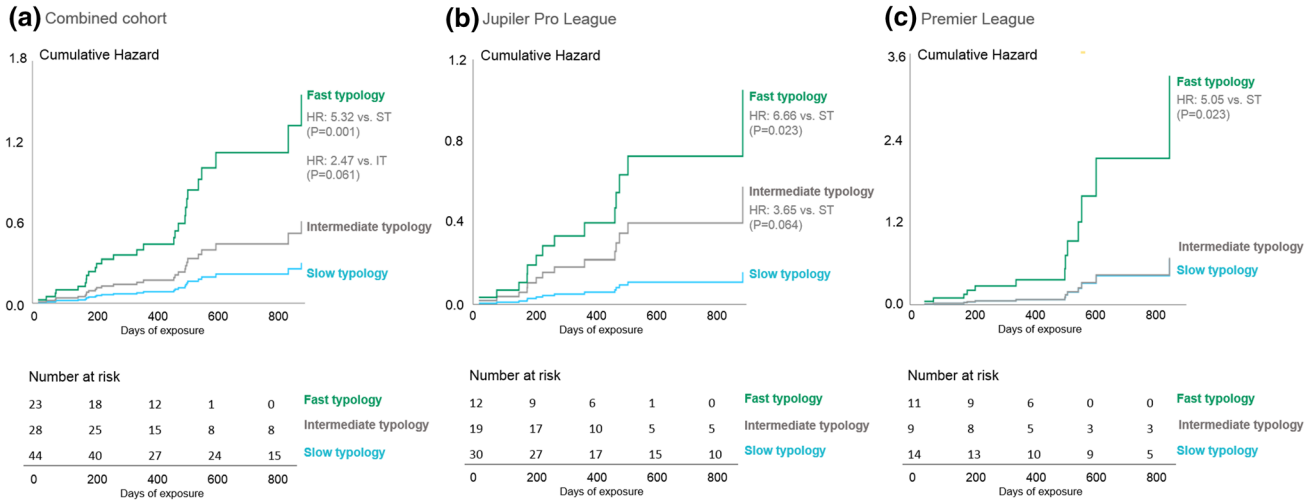

Fig. 4 Cumulative hazard curves and number at risk in **a** combined cohort, **b** Jupiler Pro League, and **c** Premier League. *HR* hazard ratio, *IT* intermediate typology, *ST* slow typology

higher than the previously proposed risk factors such as higher age (HR 1.1) and hamstring injury history (HR 3.5) [27]. Moreover the sensitivity and specificity, described by the area under the ROC curve (AUC 0.67–0.80), were comparable with eccentric knee flexor torque and fascicle length (AUC 0.65, relative risk [RR] 4.4 and AUC 0.71, RR 4.1, respectively) [5].

It is generally acknowledged that a large portion of the biopsy-based muscle fibre typology is genetically determined [28]. A previous twin study that used the same non-invasive estimate of muscle fibre typology likewise revealed an 85% heritability (95% CI 0.71–0.91) for the soleus carnosine content [29]. Whether or not this implies that the increased risk for HSI in FT players is genetically determined and therefore inevitable depends on the potential causal mechanism of the relationship. We propose two explanations that would render the muscle fibre typology either a non-modifiable or a modifiable risk factor.

First, the higher vulnerability of FT players may be related to structural differences between fast- and slow-twitch muscle fibres [30]. Fast-twitch fibres contain narrower Z-discs, smaller isoforms of nebulin and myomesin, and less elastic isoforms of the protein titin, which results in a lower sarcomeric integrity when compared with slow-twitch fibres. Moreover, dystrophin, which connects the cytoskeleton to the sarcolemma, has a twofold lower content in fast-twitch fibres than in slow-twitch fibres. Also, the frequency of sarcomere branching, providing a pathway for lateral and longitudinal active force transmission, is 2.8-fold lower in fast-twitch than slow-twitch fibres [31]. Additionally, fast-twitch fibres can produce higher power than slow-twitch fibres, which results in an even larger imbalance between the higher load and the lower load capacity. In vitro studies showed that these factors make fast-twitch fibres inherently more vulnerable to strain-induced damage [30].

A second potential reason for the higher HSI risk in FT players is based on the link between fatigue and HSI. In a previous study from our laboratory, a priori identified FT subjects showed a higher fatigue accumulation during three Wingate tests and took 15 times as long to recover from this high-intensity exercise than those in the ST group [32]. When generalising these data to football players, FT players might accumulate fatigue during their games, possibly resulting in a higher injury susceptibility. More specifically, it is suspected that hamstring injuries are mostly sustained during the terminal swing phase of high-intensity running, in which the hamstrings have to work eccentrically to control the knee extension force. If hamstring-related fatigue during these eccentric actions is predominantly present in FT players, their hamstrings might no longer be able to produce the high forces needed to control the knee extensions, possibly resulting in overlengthening and tears [33, 34].

The extent to which these potential mechanisms (structural weakness and/or fatigue) contribute to our findings and whether they would work synergistically remains to be determined. However, in contrast to structural weakness, the higher fatigability might be partly modifiable by individualising training and recovery cycles. While most team players nowadays follow a one-size-fits-all training regimen, FT players may be disadvantaged and could instead benefit from individualised strategies for their training–recovery cycles. We believe that individualisation based on player position (keeper/defender/midfielder/forward) will not meet these demands, as muscle typologies were somewhat equally distributed over all player positions in our study (see Fig. 3). Therefore, non-invasive identification of muscle fibre typology groups could be considered for more appropriate individualisation of training prescription and injury-prevention strategies.

We acknowledge that the muscle fibre typology was non-invasively estimated in the soleus and not in the hamstrings, which can be seen as a limitation. However, given the presence of an across-muscle phenotype, we believe that the muscle fibre typology of the hamstrings is related to the Z-score measured in the soleus muscle [21, 22]. Moreover, the two different MRI systems used to measure the carnosine content in the soleus introduced differences in the acquisition and in the reference population between sites. The relatively low number of HSI in the Jupiler Pro League ($n = 14$) and the Premier League ($n = 13$) introduced the possibility of sparse data bias [35] and resulted in a low fragility index [36]; therefore, results should be interpreted with caution. However, the different study locations enabled us to study larger groups and test the reproducibility of the findings, which increased their power.

5 Conclusion

Muscle fibre typology has been revealed as a novel risk factor for HSI in team sports, with FT players displaying a 5.3-fold higher risk of experiencing a new HSI than ST players. This risk may be partly modifiable when fatigue is involved in this relationship. Future research should use a multifactorial design to highlight the interrelatedness and importance of all established risk factors in an attempt to lower the incidence of strains in the future.

Declarations

Funding EL received a grant from Research Foundation-Flanders (FWO 1104020 N) during the conduct of the study. The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Conflict of interest EL received a grant from Research Foundation-Flanders (FWO 1104020 N) during the conduct of the study. KVV, FVdC, EW, DD, BDW, SM, WDG, JBM, JGB, EW, and WD have no conflicts of interest that are directly relevant to the content of this article.

Ethics approval This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the local ethical committee (Ghent University Hospital, Ghent, Belgium; B670201837890).

Consent to participate All subjects gave their written informed consent to take part in the study.

Consent for publication Not applicable.

Availability of data and material The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability Not applicable.

Author contributions EL, WD, and EW participated in the design of the study and drafted the manuscript. KVV, FVdC, and EW contributed to the statistical analysis. DD, SM, BDW, SM, WDG, JBM, and JGB participated in data collection. All authors contributed to data interpretation and reviewed the final version.

References

1. Ekstrand J, Hägglund M, Waldén M. Epidemiology of muscle injuries in professional football (soccer). Am J Sports Med. 2011;39:1226–32.
2. Petersen J, Hölmich P. Evidence based prevention of hamstring injuries in sport. Br J Sports Med. 2005;39:319–23.
3. Ekstrand J, Waldén M, Hägglund M. Hamstring injuries have increased by 4% annually in men's professional football, since 2001: a 13-year longitudinal analysis of the UEFA Elite Club injury study. Br J Sports Med. 2016;50:731–7.
4. Freckleton G, Pizzari T. Risk factors for hamstring muscle strain injury in sport: a systematic review and meta-analysis. Br J Sports Med. 2013;47:351–8.
5. Timmins RG, Bourne MN, Shield AJ, Williams MD, Lorenzen C, Opar DA. Short biceps femoris fascicles and eccentric knee flexor weakness increase the risk of hamstring injury in elite football (soccer): a prospective cohort study. Br J Sports Med. 2016;50:1524–35.
6. Barnes C, Archer DT, Hogg B, Bush M, Bradley PS. The evolution of physical and technical performance parameters in the english premier league. Int J Sports Med. 2014;35:1095–100.
7. Woods C, Hawkins RD, Maltby S, Hulse M, Thomas A, Hodson A. The football association medical research programme: an audit of injuries in professional football—analysis of hamstring injuries. Br J Sports Med. 2004;38:36–41.
8. Huygaerts S, Cos F, Cohen DD, Calleja-González J, Guitart M, Blazevich AJ, et al. Mechanisms of hamstring strain injury: interactions between fatigue. Muscle Act Funct Sports. 2020;8:65.
9. Schuermans J, Van Tiggelen D, Danneels L, Witvrouw E. Biceps femoris and semitendinosus—teammates or competitors? New insights into hamstring injury mechanisms in male football players: a muscle functional MRI study. Br J Sports Med. 2014;48:1599–606.
10. Small K, McNaughton L, Greig M, Lovell R. The effects of multidirectional soccer-specific fatigue on markers of hamstring injury risk. J Sci Med Sport. 2010;13:120–5.
11. Allen TJ, Leung M, Proske U. The effect of fatigue from exercise on human limb position sense. J Physiol. 2010;588:1369–77.
12. Morgan DL. New insights into the behavior of muscle during active lengthening. Biophys J Cell Press. 1990;57:209–21.
13. Lehnert M, De Ste CM, Zaatar A, Hughes J, Varekova R, Lastovicka O. Muscular and neuromuscular control following soccer-specific exercise in male youth: Changes in injury risk mechanisms. Scand J Med Sci Sport. 2017;27:975–82.
14. Greig M. The influence of soccer-specific fatigue on peak isokinetic torque production of the knee flexors and extensors. Am J Sports Med. 2008;36:1403–9.
15. Timmins RG, Opar DA, Williams MD, Schache AG, Dear NM, Shield AJ. Reduced biceps femoris myoelectrical activity influences eccentric knee flexor weakness after repeat sprint running. Scand J Med Sci Sport. 2014;24:299–305.
16. Greco CC, Da Silva WL, Camarda SRA, Denadai BS. Fatigue and rapid hamstring/quadriceps force capacity in professional soccer players. Clin Physiol Funct Imaging. 2013;33:18–23.
17. Thorstensson A, Karlsson J. Fatiguability and fibre composition of human skeletal muscle. Acta Physiol Scand. 1976;98:318–22.
18. Baguet A, Everaert I, Hespel P, Petrovic M, Achten E, Derave W. A new method for non-invasive estimation of human muscle fiber type composition. PLoS ONE. 2011;6:e21956.
19. Ozdemir MS, Reyngoudt H, De Deene Y, Sazak HS, Fieremans E, Delputte S, et al. Absolute quantification of carnosine in human calf muscle by proton magnetic resonance spectroscopy. Phys Med Biol. 2007;52:6781–94.
20. Baguet A, Reyngoudt H, Pottier A, Everaert I, Callens S, Achten E, et al. Carnosine loading and washout in human skeletal muscles. J Appl Physiol. 2009;106:837–42.
21. Vikne H, Gundersen K, Liestøl K, Maelen J, Vøllestad N. Intermuscular relationship of human muscle fiber type proportions: Slow leg muscles predict slow neck muscles. Muscle Nerve. 2012;45:527–35.
22. Johnson MA, Polgar J, Weightman D, Appleton D. Data on the distribution of fibre types in thirty-six human muscles. An autopsy study. J Neurol Sci. 1973;18:111–29.
23. Baguet A, Bourgois J, Vanhee L, Achten E, Derave W. Important role of muscle carnosine in rowing performance. J Appl Physiol. 2010;109:1096–101.
24. MacMillan EL, Bolliger CS, Boesch C, Kreis R. Influence of muscle fiber orientation on water and metabolite relaxation times, magnetization transfer, and visibility in human skeletal muscle. Magn Reson Med. 2016;75:1764–70.
25. Just Kukurová I, Valkovič L, Ukropec J, de Courten B, Chmelík M, Ukropcová B, et al. Improved spectral resolution and high reliability of in vivo 1H MRS at 7 T allow the characterization of the effect of acute exercise on carnosine in skeletal muscle. NMR Biomed. 2016;29:24–32.
26. Lievens E, Van Vossel K, Van de Casteele F, Krssak M, Murdoch JB, Befroy DE, et al. Cores of Reproducibility in Physiology (CORP): quantification of human skeletal muscle carnosine concentration by proton magnetic resonance spectroscopy. J Appl Physiol. 2021. (**Epub ahead**)
27. Hägglund M, Waldén M. Previous injury as a risk factor for injury in elite football: a prospective study over two consecutive seasons. Br J Sport Med. 2006;40:767–72.
28. Simoneau JA, Bouchard C. Genetic determinism of fiber type proportion in human skeletal muscle. FASEB J. 1995;9:1091–5.
29. Baguet A, Everaert I, Achten E, Thomis M, Derave W. The influence of sex, age and heritability on human skeletal muscle carnosine content. Amino Acids. 2012;43:13–20.

30. Qaisar R, Bhaskaran S, Van Remmen H. Muscle fiber type diversification during exercise and regeneration. Free Radic Biol Med. 2016;98:56–67.
31. Willingham TB, Kim Y, Lindberg E, Bleck CKE, Glancy B. The unified myofibrillar matrix for force generation in muscle. Nat Commun. 2020;11:1–10.
32. Lievens E, Klass M, Bex T, Derave W. Muscle fiber typology substantially influences time to recover from high-intensity exercise. J Appl Physiol. 2020;128:648–59.
33. Chumanov ES, Schache AG, Heiderscheit BC, Thelen DG. Hamstrings are most susceptible to injury during the late swing phase of sprinting. Br J Sports Med. 2012;46:90–90.
34. Kenneally-Dabrowski CJB, Brown NAT, Lai AKM, Perriman D, Spratford W, Serpell BG. Late swing or early stance? A narrative review of hamstring injury mechanisms during high-speed running. Scand J Med Sci Sport. 2019;29:1083–91.
35. Greenland S, Mansournia MA, Altman DG. Sparse data bias: a problem hiding in plain sight. BMJ. 2016;353.
36. Walsh M, Srinathan SK, McAuley DF, Mrkobrada M, Levine O, Ribic C, et al. The statistical significance of randomized controlled trial results is frequently fragile: a case for a Fragility Index. J Clin Epidemiol. 2014;67:622–8.